

MANN 192311090

CO3 AT1



SIMATS Online Programming Lab
JAVA PROGRAMMING

MANN 192311090RB

QUESTIONS

Q1
ArrayIndexOutOfBoundsException
10 marks

Q2
calculator application
10 marks

Q3
student registration system
10 marks

3/3 answered

Q2 calculator application 10 marks

You are building a simple calculator application. Write a program that takes two integers as input and divides them. Handle the case where the denominator is zero.

Your Code

```
1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int a = sc.nextInt(), b = sc.nextInt();
6         try {
7             System.out.println(a / b);
8         } catch (ArithmeticException e) {
9             System.out.println("Cannot divide by zero");
10        }
11    }
12 }
```

Input

20 0

Output

Run

Save



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3/3 answered

Q1 ArrayIndexOutOfBoundsException 10 marks

You are developing a student grade management system. Write a program that stores 5 student grades in an array and allows users to retrieve a grade by entering the student's position (1-5). Handle cases where the position is out of range or input is invalid
Sample Output : 15

Your Code

```
1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         int[] grades = {85, 90, 78, 92, 60};
5         Scanner sc = new Scanner(System.in);
6         try {
7             int pos = Integer.parseInt(sc.nextLine().trim());
8             System.out.println(grades[pos - 1]);
9         } catch (ArrayIndexOutOfBoundsException e) {
10            System.out.println("Invalid position!");
11        } catch (NumberFormatException e) {
12            System.out.println("Invalid input! Enter a valid number.");
13        }
14    }
15 }
```

Input

3

Output

QUESTIONS

Q1
Student Registration System
10 marks

Q2
E-Commerce Cart System
10 marks

Q3
Library Management System
10 marks

Q4
Employee Data Processing
10 marks

Q5
Online Voting System
10 marks

5/5 answered

Q1 Student Registration System

10 marks

Suppose that a university system stores student data, where each student must be unique and can enroll in multiple courses.

(a) Enumerate three Collection interfaces suitable for this system.

(b) Design a Collection structure (diagram/representation) to store students and their enrolled courses.

(c) Write a Java program using HashSet and HashMap with Generics to:

- Add students
- Map students to courses
- Display data using Iterator

Your Code

```
1 import java.util.*;
2
3 public class Main {
4     public static void main(String[] args) {
5         HashSet<Student> set = new HashSet<>();
6         HashMap<Student, HashSet<String>> map = new HashMap<>();
7         Student s1 = new Student("S1", "MANN");
8         Student s2 = new Student("S2", "RAVI");
9         set.add(s1);
10        set.add(s2);
11        HashSet<String> c1 = new HashSet<>();
12        c1.add("Java");
13        c1.add("DBMS");
14        HashSet<String> c2 = new HashSet<>();
```

Input

stdin input

Output